20 · The three layers of the eye

The words of this track, as it says them. Bold brown is the examiner.

The three layers of the eye. Four and a half marks.

What are the three layers, or tunics, of the eye? Give their structures and their functions.

The hook, from the outside in: the tough coat, the blood coat, and the seeing coat.

The tough coat is the fibrous tunic, on the outside. It has two parts. The white sclera, which protects the eye and anchors the muscles that move it. And the clear cornea at the front, which lets light in, and does about eighty per cent of the refraction.

The blood coat is the vascular tunic, also called the uvea, in the middle. It has three parts. The choroid: pigmented and rich in blood vessels. It nourishes the other layers, and stops light scattering. The ciliary body: its ciliary muscles and suspensory ligaments hold the lens and change its shape, and its processes make the aqueous humour. And the iris: its circular and radial muscles control the size of the pupil, and so how much light gets in.

The seeing coat is the neural tunic, the retina, on the inside. Its outer pigmented layer absorbs light and stops it scattering. Its inner neural layer has three layers of neurons: the photoreceptors, rods and cones, then bipolar cells, then ganglion cells, whose axons leave the eye as the optic nerve, cranial nerve two. The photoreceptors turn light into nerve impulses.

The traps. The cornea, not the lens, does most of the refraction: about eighty per cent. Second: the choroid belongs to the middle, vascular tunic, not the retina. Third: the rods and cones are in the neural layer. The pigmented layer only absorbs light.

Once more. The three layers of the eye.

The fibrous tunic: the sclera, which protects the eye and anchors its muscles, and the cornea, which does about eighty per cent of the refraction. The vascular tunic: the choroid, rich in blood vessels, which nourishes the eye; the ciliary body, which holds the lens, changes its shape, and makes aqueous humour; and the iris, which controls pupil size. The neural tunic, the retina: a pigmented layer that absorbs light, then photoreceptors, rods and cones, then bipolar cells, then ganglion cells, whose axons form the optic nerve.